A Fine Extraction Method for Farmland Plots Based on Remote Sensing With Semantic and Edge Feature Fusion

Kaixin Kong, Xiaoxia Yang, Chengming Zhang, Jinlong Fan, Feng Li, Menxin Wu, Xuehui Hou, and Lin Ding

***Abstract*—Accurate agricultural parcel data serves as crucial foundational information in modern agriculture. Addressing the urgent demand for fine-grained extraction of agricultural parcels from high-resolution remote sensing imagery, this article proposes a fine extraction method for farmland plots based on remote sensing with semantic and edge feature fusion (FFENet). FFENet takes high-resolution remote sensing imagery as input and mainly comprises core components such as a semantic feature encoding module, an edge feature encoding module (EEM), a hybrid attention-based edge attention module (HA-EAM), a feature fusion module (FFM), a feature generation module, and a classification module (EDWC). Specifically, the FFM is used for hierarchical fusion of semantic features and edge features, the HA-EAM uses edge features to predict potential parcel boundaries, and the feature generation module decodes the output of the FFM to generate pixelwise features. Finally, EDWC integrates the pixelwise features and their probability of being a boundary to determine the parcel classification of each pixel. To verify the effectiveness and generalization capability of the proposed method, this study selected Taian City and Juye County of Shandong Province as research areas and used Gaofen-2 (GF-2) PMS imagery as the data source for the experiments. Comparative results against mainstream models such as DeeplabV3+, Swin-Unet, and edge information guided network (EIGNet) demonstrated that FFENet achieved optimal performance in the parcel extraction task. The results showed that FFENet achieved an *F*1-score of 94.6% and an IoU of 89.5% on this dataset, significantly outperforming the comparison models. The proposed approach exhibits superior performance in boundary extraction accuracy and shape consistency, proving its significant advantages in the fine-grained extraction of agricultural parcels.**

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I. Introduction

AGRICULTURAL parcels, as the basic unit of agricultural production, are crucial foundational elements for the digital and intelligent development of modern agriculture [1]. In critical scenarios such as agricultural insurance claims assessment, farmland protection monitoring, subsidy distribution, and smart agriculture scheduling, the precision of parcel boundaries and the consistency of their shape directly determine the reliability of data applications [2]. Although current parcel extraction studies have achieved relatively high overall accuracy (OA), literature shows that extracted parcel shapes still deviate from field structures, particularly in boundary completeness and corner sharpness, severely restricting their practical utility [3]. Therefore, extracting fine-grained parcel information with accurate boundaries and realistic shapes from high-resolution remote sensing imagery has become an important research topic in agricultural remote sensing.

Pixelwise classification of high-resolution remote sensing imagery serves as the foundation for acquiring parcel data [4], whereas high-quality feature representation is central to effective image segmentation. Previous research relied primarily on basic features such as pixel spectral values [5], [6], [7], textural features [8], [9], and normalized difference vegetation index [10]. These features were combined with traditional machine learning algorithms, including thresholding [11], *K*-means algorithm [12], and random forest [13], to achieve remote sensing image segmentation, yielding good results on medium-to-low-resolution imagery [14]. However, high-resolution remote sensing imagery features complex scenes and rich details [15], making it difficult for these traditional methods to cope with complex background interference, thus gradually failing to meet the demand for fine-grained extraction.

Convolutional neural networks (CNNs) [16] have emerged as key technologies to address these challenges because of their powerful feature learning capabilities. Fully convolutional networks [17] first enabled end-to-end semantic segmentation. Subsequently, models such as UNet [18], [19], SegNet [20], [21], ResNet [22], [23], RefineNet [24], [25], and ERFNet [26] were proposed and applied to parcel extraction, significantly enhancing segmentation accuracy. However, these networks tend to emphasize the extraction of

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homogeneous semantic features within parcels during encoding, while responding insufficiently to geometric boundaries between parcels. This often leads to errors in corners and contours, ultimately resulting in distorted parcel shapes.

Agricultural parcels also contain important structural priors: field interiors typically exhibit homogeneity in color and texture, whereas clear physical boundaries separate different fields [27], [28]. Traditional edge detection operators, such as Canny [29] and Sobel [30], capture such boundaries by computing gradients via image filtering, but they face significant difficulties in extracting boundaries from complex agricultural remote sensing imagery [31]. First, a single gradient threshold cannot adapt to variations in illumination and crop types across an entire image, limiting generalizability [32]. Second, parcel sizes vary greatly, complicating multiscale boundary detection; boundaries of small parcels are especially susceptible to high-frequency noise [33]. These limitations are mainly due to the lack of semantic understanding of image content in traditional edge detection operators, resulting in breaks, artifacts, and spurious edges upon direct application, making it difficult to effectively distinguish true parcel boundaries from other types of edge interference [34].

Nevertheless, edge information carries rich geometric knowledge, providing valuable clues for enhancing the segmentation performance of deep learning models. Therefore, effectively leveraging the robustness of traditional operators while compensating for their lack of semantic understanding has become a key research direction. Studies have shown that using edge information as a prior constraint can effectively guide the model to focus on field boundary regions, alleviate semantic ambiguity, and thus improve the accuracy of segmentation boundaries [35], [36].

Building on the effectiveness of traditional edge detection methods as auxiliary information, researchers have further focused on how to more effectively integrate edge information into deep neural network architectures [37], [38], aiming to specifically address the misclassification issue at the corners of ground objects. For example, [39] proposed the edge information guided network (EIGNet), which enhances segmentation accuracy via a directional convolution module and a spatial–semantic feature aggregation module. Lu et al. [40] developed REAUNet, which strengthens the edge extraction capability of farmland parcels through deep supervision, an edge detection module, and a dual attention mechanism, thereby significantly improving segmentation accuracy and model transferability. Zhao et al. [41] proposed HBGNet, which enhances boundary perception using Laplacian convolution, and constructed China's first large-scale high-resolution farmland plot dataset, exhibiting markedly enhanced plot extraction accuracy across multiple scenarios. Zhu et al. [42] proposed a highly generalizable multitask detection network coupled with a plot optimization step by integrating boundary and region dual-task decoders, color space augmentation, and a hierarchical attention mechanism. The network significantly improved both the mapping accuracy and spatiotemporal generalization ability of cultivated land plots.

The quality of the fine-field extraction depends on both the neural network's feature extraction capability and the classifier's discriminative power. These two abilities are mutually reinforcing, and optimal results require their simultaneous and coordinated enhancement [43]. However, most existing research emphasizes improving feature extraction, with relatively less attention to classifier design.

Recently, researchers have focused on the critical role of classifiers in image segmentation. Several new pixel-level classifiers based on the softmax classifier [44] have been proposed. For instance, StackNet combines different classifiers to enhance performance [45], while regionwise loss addresses category imbalance and pixel importance, fine-tuning the softmax classifier to boost accuracy [43]. Although these classifiers improve classification performance to some extent, they still rely heavily on the inherent pixel information for classification [46], [47], limiting their use of contextual relationships. This shortcoming is particularly problematic for extracting faceted objects like fields, as these classifiers struggle to capture relationships between pixels and object boundaries [48]. Consequently, extracted field shapes often differ considerably from the actual geometry, hindering applications such as agricultural insurance [49].

In response to the urgent demand for fine-grained extraction of agricultural parcels from high-resolution remote sensing imagery, we propose a fine extraction method for farmland plots based on remote sensing with semantic and edge feature fusion (FFENet). Compared with traditional CNNs, FFENet fully uses local and global semantic and edge features, significantly enhancing the partitioning accuracy of agricultural parcels. This proposed method is developed and evaluated using typical farmland parcel data.

The core contributions of this study are threefold: 1) we design a pixelwise feature extraction framework that integrates edge and semantic features for fine-grained parcel representation; 2) we propose an edge-aware classification method that improves boundary and corner discrimination; and 3) we develop an edge-enhanced loss function that strengthens boundary supervision and improves parcel shape fidelity.

II. Study Area and Data

A. Study Area

To verify the effectiveness and generalization capability of the proposed agricultural parcel extraction method, this study selected Taian City and Juye County of Shandong Province as research areas.

The Taian City study area is situated in the transitional zone between the East China Plain and the central mountainous area of Shandong Province. Its diverse geomorphic types, encompassing plains and hills, offer a variety of typical farmland landscapes, which provides a basis for model training and robustness evaluation under different topographic conditions. We selected the winter wheat fields in this area as the main research object because their field interiors exhibit a high degree of spectral uniformity during the experimental period, and they show a significant contrast with boundary objects such as field ridges and roads, facilitating the construction of high-precision annotated data.

The Juye County research area is situated in the East China Plain, featuring relatively flat and regular parcels, and we selected the garlic fields in this area as the research object. The garlic fields present a significant contrast to the winter wheat fields in terms of plant morphology, canopy structure, and

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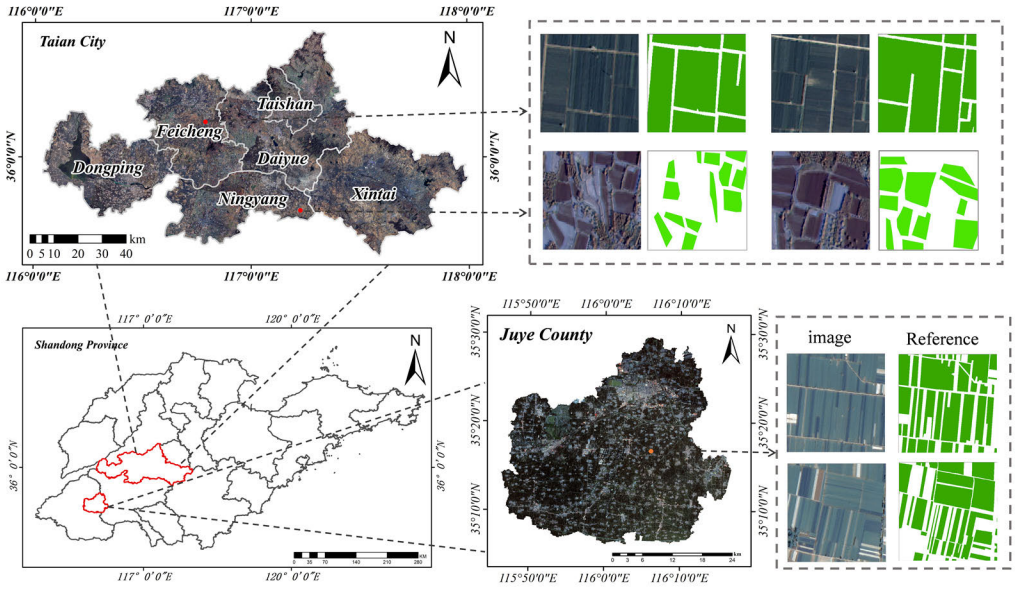


Fig. 1. Location map, remote sensing image, and sample types of the research area.

parcel geometric structure. This comparative validation system can not only evaluate the model's robustness under different geomorphic conditions but also comprehensively assess the generalization capability of the geometric structure priors learned by FFENet for cross-crop parcel extraction.

B. Data and Preprocessing

This study used Gaofen-2 (GF-2) PMS imagery as the data source, selecting 42 scenes acquired between March and April 2024. Through preprocessing steps including radiometric calibration, atmospheric correction, orthorectification, and image fusion, 1-m resolution images containing blue, green, red, and near-infrared bands were generated. As shown in Fig. 1, the distribution of the study area, the remote sensing imagery, and the sample types are illustrated.

We conducted a joint ground survey with the Meteorological Bureau, obtaining a total of 560 survey points (300 for parcels and 260 for other types). Taian City's main surface types include buildings, bare soil, water bodies, roads, forestland, and cultivated fields. Leveraging the distinct differences between the cultivated fields and other features in the GF-2 imagery, marking was performed via visual interpretation, setting 1 as the attribution class for parcel pixels and 0 for nonparcel pixels. Finally, the remote sensing images and corresponding annotated images were segmented into 512×512-pixel blocks, yielding 1800 pairs of "remote sensing image–annotated image" to construct the training and testing datasets for the model.

III. Methods

The overall approach comprises four key components: designing the model structure, defining the loss function, conducting model training and testing, and performing feature extraction. The basic framework of this approach is illustrated in Fig. 2.

A. FFENet

FFENet takes high-resolution remote sensing images as input and comprises several key components, including a semantic feature encoding module, the EEM, the feature fusion module (FFM), the hybrid attention-based edge attention module (HA-EAM), and the EDWC. The overall structure of the model is illustrated in Fig. 3.

1) Semantic Feature Encoding Module: Owing to the significant scale variation in winter wheat fields and the increasing resolution of remote sensing images, the spatial details and complexity of target features have grown, making it challenging for conventional shallow networks to extract effective features. To address this issue and enable deeper network construction while avoiding the vanishing gradient problem, we used the ResNet architecture for feature extraction from winter wheat fields. Specifically, we selected ResNet18 as the backbone for semantic feature extraction, with the global average pooling and fully connected layers discarded. Compared with standard convolutional networks, residual-structured convolutional networks can extract more complex image features while effectively mitigating issues such as gradient vanishing that may arise as the number of convolutional layers increases. The structural diagram and detailed parameter settings of the network are shown in the box on the left side of Fig. 3.

2) EEM: As shown in Fig. 3, the EEM comprises four blocks. Each block incorporates an improved Canny algorithm that integrates edge gradient information during the feature-extraction process, thereby enhancing the model's ability to recognize field boundaries more accurately. The traditional Canny algorithm uses the Sobel operator as the gradient operator and calculates the gradients in the horizontal and vertical directions by applying first-order finite differences to the filtered image. Building on this, our method retains the Sobel operator for horizontal and vertical gradient detection while also introducing additional gradient calculations in two diagonal directions (45° and 135°). This

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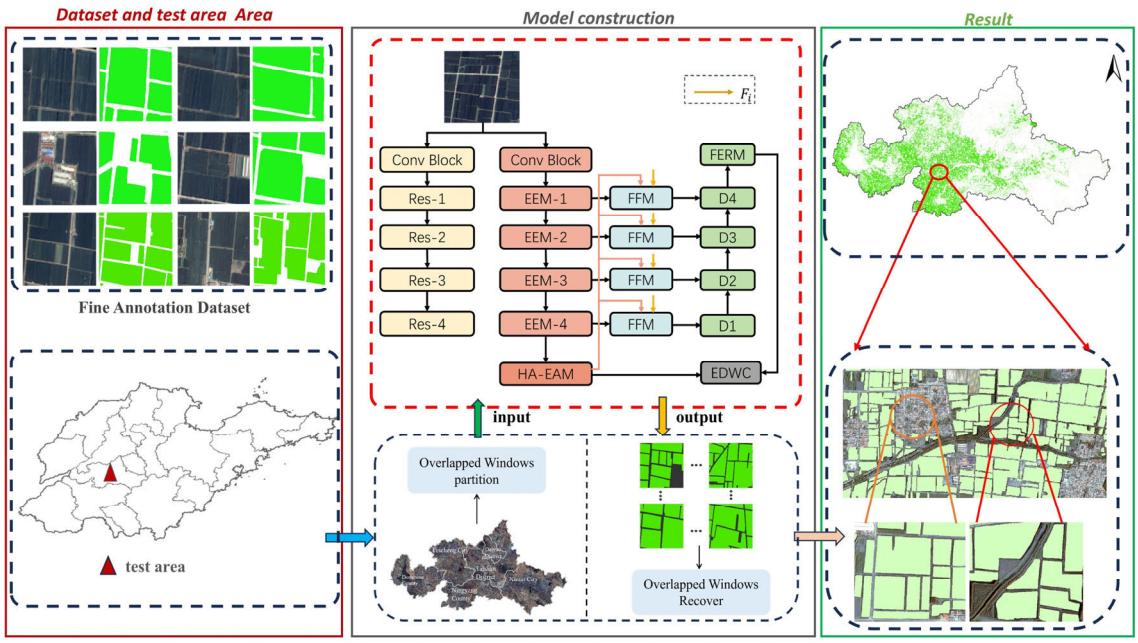


Fig. 2. Overview of the proposed mapping framework. The framework comprises three key components. (Left) Model training. (Middle) Data processing. (Right) Model inference.

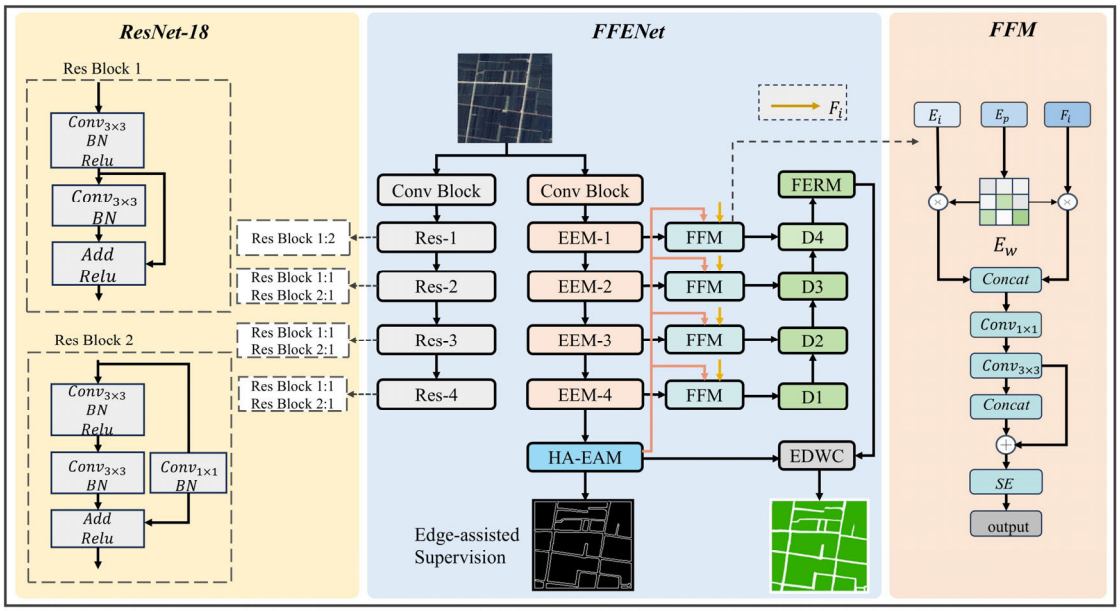


Fig. 3. Structure of FFENet model (overall framework with res-n and HA-EAM modules). Res-n: semantic encoder; EEM-n: edge encoder; HA-EAM: edge-aware module; D-n: upsampling; FERM: refinement; EDWC: classifier.

enhancement allows the detection of edge features from all four directions, thus improving edge extraction performance. FFENet further enhances the model by integrating a traditional edge-detection operator. By leveraging this prior knowledge, the model's ability to extract complex semantic features from a limited number of samples was significantly enhanced. In addition, the gradient features from all directions are fused using a convolutional layer, while residual connections are used to mitigate potential edge feature omissions. The detailed computational process is shown in Fig. 4(a).

3) Hybrid Attention-Based Edge Attention Module: To further enhance the efficiency of field-edge prediction, this study introduces an edge-aware module (HA-EAM) based on an attention mechanism. As shown in Fig. 5, a hybrid attention mechanism was used to progressively compress the input feature maps along both the channel and spatial dimensions. The extracted feature weights were then used to recalibrate the initial feature maps and improve the feature representation. The detailed process is illustrated in the figure.

The HA-EAM leverages the feature maps $\{E_i, i = 2, 3, 4\}$ extracted by the EEM for edge prediction. First, the feature

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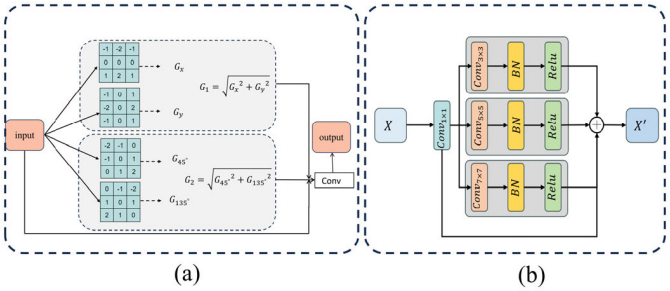


Fig. 4. Structures of the EEM and FERM. (a) Structure of the EEM module and (b) structure of the FERM module.

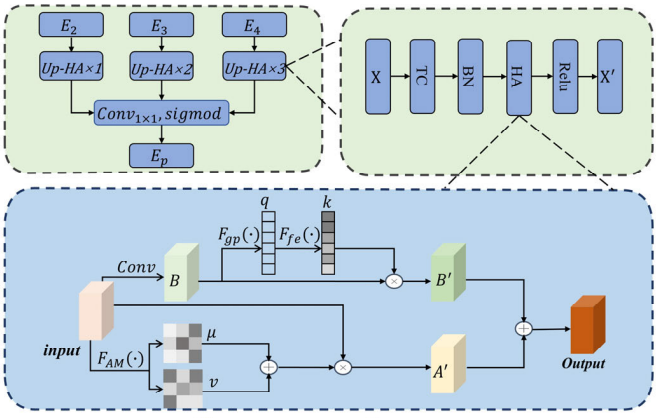


Fig. 5. Structure of the HA-EAM module.

maps were upsampled to align their resolutions with those of the input image. Specifically, E_2, E_3, and E_4 underwent two-, four-, and eightfold upsampling, respectively. The upsampling process is depicted by the black dashed box on the top right of Fig. 5. Subsequently, hybrid attention was applied to further refine the edge features from both the spatial and channel perspectives. Finally, the refined features were concatenated and passed through a 1×1 convolutional layer to output the final edge-prediction map, E_p.

4) Feature Fusion Module: The shallow features of ResNet18 contain rich spatial information, whereas the deeper layers capture more abstract semantic information. Moreover, the features generated by the EEM retained significant spatial details. Directly fusing these diverse features without accounting for their inherent differences may negatively impact model performance. To address this issue, we proposed a FFE based on an attention mechanism and a residual structure.

A schematic of the FFM is shown in the box on the right side of Fig. 3. The FFM accepted two input features of identical scale from the semantic feature branch, the spatial feature branch, and the edge weights. First, the edge weight map E_w was derived from the edge prediction map E_p, as shown in (1). Semantic feature map F_i and spatial edge feature map E_i were then weighted individually, and the resulting weighted feature maps were concatenated. Next, the intermediate results were generated using 1×1 convolutional layers. These intermediate results were fed into a residual structure that incorporated a channel attention mechanism that learns the correlations between the input feature channels and emphasizes the important feature channels by assigning them higher weights

$$E_w = \text{sigmoid}\left(\text{Conv}_{1\times1}\left(\text{Avgpool}\left(E_p\right)\right)\right). \tag{1}$$

5) Feature Generation Module: As shown in Fig. 3, the module adopts a cascade decoding structure that combines upsampling with feature refinement enhancement to restore the resolution of the extracted remote sensing image features.

The feature-generation module comprised four decoding units (D1–D4) and one feature-refinement unit. Each decoding unit received two inputs: the feature output from the FFM module and the upsampled features from the previous decoding unit (D1 only receives input from the FFM). The high-level decoded features were first upsampled to match the resolution of the FFM fusion features, after which the two feature maps were channel-cascaded to enable cross-layer feature stitching. To extract more precise feature information, we incorporated a feature refinement module (FERM) into the model. As shown in Fig. 4(b), the feature-refinement unit consists of a convolutional layer and a residual unit. Convolutional layers with sizes of 3×3, 5×5, and 7×7 in the residual unit correspond to three refinement operations with different receptive fields. This allows the module to extract additional feature information while preventing pixel loss in the feature map caused by direct upsampling, thereby refining the output features to a certain extent.

6) Classification Module: Fig. 6 shows the structural diagram of the EDWC. As previously mentioned, in remote sensing images, the probability of pixel misclassification inside a parcel is very low, and pixels can be accurately classified based on their features. However, pixels located at the edges must leverage not only their semantic features but also the semantic and edge features of the surrounding pixels to be correctly classified. To address this, FFENet uses edge density to represent the number of edge pixels within an image block centered around the pixel to be classified. Specifically, edge density is defined as the ratio of the number of edge pixels within an image block centered around the pixel to the area of the image block. This definition is expressed as

$$\text{ED} = \frac{S}{W \times H} = \frac{\sum_{i=1}^{W}\sum_{j=1}^{H} e\left(x, y\right)}{W \times H} \tag{2}$$

where ED denotes the edge density; S denotes the sum of the edge intensity values in the region; W and H denote the width and height of the image block, respectively; and $e(x, y)$ denotes the edge map.

Higher edge density indicates richer edge information within a region. Building on this, FFENet proposes the concept of an edge weight (EW) term to measure the degree of influence of the edge information on the classification process. It is defined by the following equation:

$$\text{EW} = \frac{E\left(x\right) \cdot e^{z_i}}{\sum_j E\left(x\right) \cdot e^{z_j}} \tag{3}$$

where $E(x)$ denotes the edge-strength value of pixel x, e^{z_i} is the exponential score of the pixel for category i, and $\sum_j e^{z_j}$ is the sum of the exponential scores for all the categories. The edge strength corrects the category probability vector output using softmax to improve classification accuracy.

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Based on this, FFENet builds a dynamically weighted classifier (EDWC) based on the normalized exponential function and the EW of the softmax model, as shown in the following equation:

$$P_{\mathrm{EDWC}}(y_i|x) = (1-\alpha)\cdot P(y_i|x) + \alpha \cdot \mathrm{EW} \tag{4}$$

where x denotes a pixel to be classified, y_i denotes some class i, $P_{\mathrm{EDWC}}(y_i|x)$ denotes the probability that x is classified into class y_i, and $P(y_i|x)$ denotes the pixel-level class probability, that is, the probability that x is labeled in class y_i, computed based on the pixel's own information. This part is computed using a softmax model, where EW denotes the effect of x-similar edge pixels on x labeled as y_i, and α denotes the weight coefficient.

$P(y_i|x)$ is defined as shown in the following equation:

$$P(y_i|x) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{5}$$

where $P(y_i|x)$ is computed using only the pixel's own feature vectors to determine the probability of the pixel belonging to a category without considering the effect of the spatial location of the pixel on the classification result, leading to potential classification errors. To fully consider the influence of pixel spatial location on the classification results, FFENet introduces edge density to express the edge information around the pixels to be classified and correct $P(y_i|x)$ and introduces the adaptive weight coefficient α to adaptively adjust $P(y_i|x)$ and EW. This design can better use edge information to correct the classification results during the classification process while avoiding the potential overfitting problem when there is insufficient edge information.

The calculation of the adaptive weight coefficient α depends on the edge density of the image. The calculation process is shown in the following equation:

$$\alpha = \max_\alpha \times (1 - \mathrm{ED}) \tag{6}$$

where $\max_\alpha$ is a preset maximum value, which needs to be determined manually during actual use. It ensures that the classifier retains a certain amount of softmax classification weights even when the edge density is low, avoiding overreliance on edge information. α can be adjusted adaptively according to the change in edge information in the image. It always varies within a reasonable range to ensure the balance of the classification process and to ensure that the classification module can flexibly handle different image types.

B. Loss Function

The loss function serves as a critical component in determining the model-training efficacy. Building on a binary cross-entropy foundation, we introduced an edge-aware loss component to formulate a joint loss function. This integrated approach enables the model to synergistically leverage both the edge features and semantic representations, significantly enhancing the pixelwise classification accuracy, particularly for challenging boundary regions, field corners, and edge transitions. The proposed loss function that drives the training of FFENet is formally defined as follows:

$$\mathrm{Loss} = \beta L_{\mathrm{BCE}} + \gamma L_{\mathrm{Edge}} \tag{7}$$

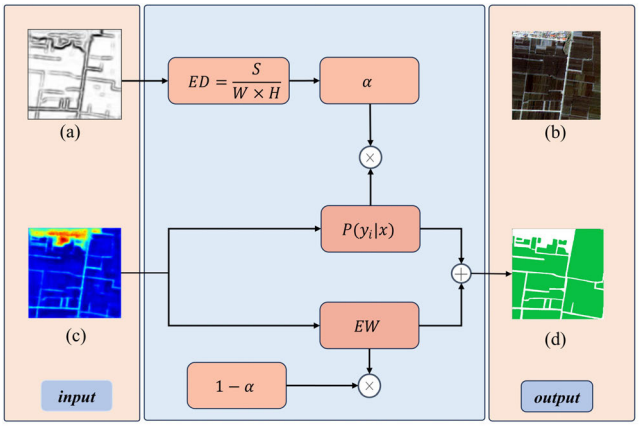


Fig. 6. Structure of the EDWC module. (a) Edge map output by the EEM, (b) feature map output by the FERM, (c) remote sensing image, and (d) final output result image.

where β and γ are the weights, L_{BCE} is the binary cross-entropy loss function, and L_{Edge} is the edge loss function. The formulas for L_{BCE} and L_{Edge} are as follows:

$$L_{\mathrm{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left(R_i \log A_i + (1-R_i)\log(1-A_i)\right) \tag{8}$$

where N is the total number of pixels, A is the predicted image segmentation result, A_i is the ith pixel in A, R is the true label value, and R_i is the ith pixel in R

$$L_{\mathrm{Edge}} = \sqrt{\sum_i^N (P_i - B_i)^2} \tag{9}$$

where N is the total number of pixels, P is the predicted image edge result, P_i is the ith pixel, B is the true edge labeling value, and B_i is the ith pixel in B.

The β and γ are hyperparameters that balance the influence of the two loss functions, whose function is to adjust the relative influence of the two tasks in gradient backpropagation.

C. Model Training and Testing

The workflow for agricultural parcel extraction using FFENet is illustrated in Fig. 7. First, the model was trained on the constructed training sample dataset and its parameters were fine-tuned using the validation sample set to achieve optimal performance. Subsequently, the optimal model was used to perform parcel feature extraction and evaluate model performance on the test areas of Taian City and Juye County of Shandong Province.

All the experiments were implemented using the PyTorch framework and trained and tested on a single NVIDIA GeForce RTX 4060Ti GPU. The dataset was split into training, validation, and test sets at a ratio of 8:1:1. Performance was evaluated based on the results of the validation set. To enhance the diversity of training data and alleviate overfitting, multiple data augmentation strategies were adopted in the preprocessing stage, including contrast adjustment, brightness adjustment, and multiangle random flipping.

The specific configuration of the experimental environment was as follows: PyTorch version 2.1.0, CUDA version 12.1,

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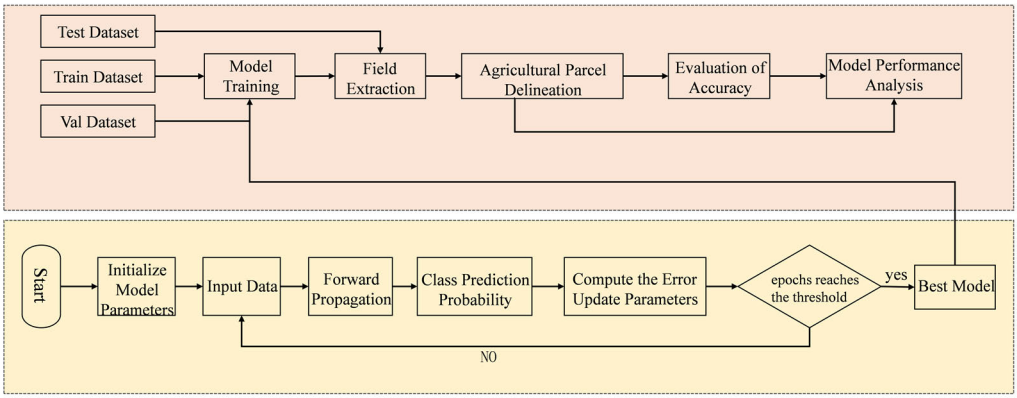


Fig. 7. Flowchart of farmland parcel distribution extraction.

TABLE I
SEMANTIC SEGMENTATION RESULTS OF FFENET AND COMPARISON MODELS ON THE TAIAN DATASET

Method	OA(%)	Precision(%)	Recall(%)	F1-Score(%)	IOU(%)
UNet	92.85	93.50	91.10	92.28	84.80
Swin-Unet	84.80	88.50	86.20	87.30	76.50
DeepLabv3+	88.87	91.95	91.00	91.47	84.05
HRNet	86.42	86.55	90.03	88.26	78.03
REAUNet	93.00	93.80	92.00	92.70	85.78
EIGNet	91.97	**95.00**	91.42	93.38	87.05
HBGNet	93.01	94.70	93.30	93.83	88.90
FFENet	**93.80**	**95.00**	**94.20**	**94.60**	**89.50**

and core dependency libraries including Python 3.9.13, GDAL 3.3.3, NumPy 1.26.4, OpenCV-Python 4.10.0.84, and Scikit-learn 1.0.2. The model uses "loss" as the loss function and the Adam optimizer as the optimization algorithm, with an input image size of 512×512. The hyperparameters used for training were as follows: the batch size was set to 4, the number of epochs was 100, and the learning rate was set to 0.005. At the end of each training epoch, a model checkpoint was saved. The model parameters achieving the lowest validation loss were selected as the best model, which was then used to evaluate the test data.

IV. EXPERIMENTS AND RESULTS

A. Comparisons to Other Methods

For a comprehensive evaluation of the effectiveness of the FFENet method, we compared it with seven representative approaches, including classical models UNet [18], DeeplabV3 + [51], Swin-UNet [50], HRNet [52], EIGNet [39], REAUNet [40], and HBGNet [41]. We retrained all the models on this dataset to ensure consistency in evaluation.

B. Evaluation Metrics

Five evaluation metrics, namely, OA, precision (P), recall (R), $F1$-score ($F1$), and intersection over union (IoU), were used to assess the test results. Their calculation formulas are as shown in the following equations:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{10}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{14}$$

C. Comparative Experiment

1) Comparative Experimental Results: Table I presents the accuracy evaluation results of FFENet and other semantic segmentation models. The proposed FFENet achieved superior performance across multiple evaluation metrics compared with other models. Specifically, FFENet attained the highest IoU and $F1$-score, at 89.50% and 94.60%, respectively, representing improvements of 13% and 11.47% over Swin-UNet and HRNet. It also achieved overall performance gains relative to mainstream models such as UNet and DeepLabv3+. Furthermore, FFENet outperformed the latest approaches, including REAUNet, EIGNet, and HBGNet, across all the evaluation

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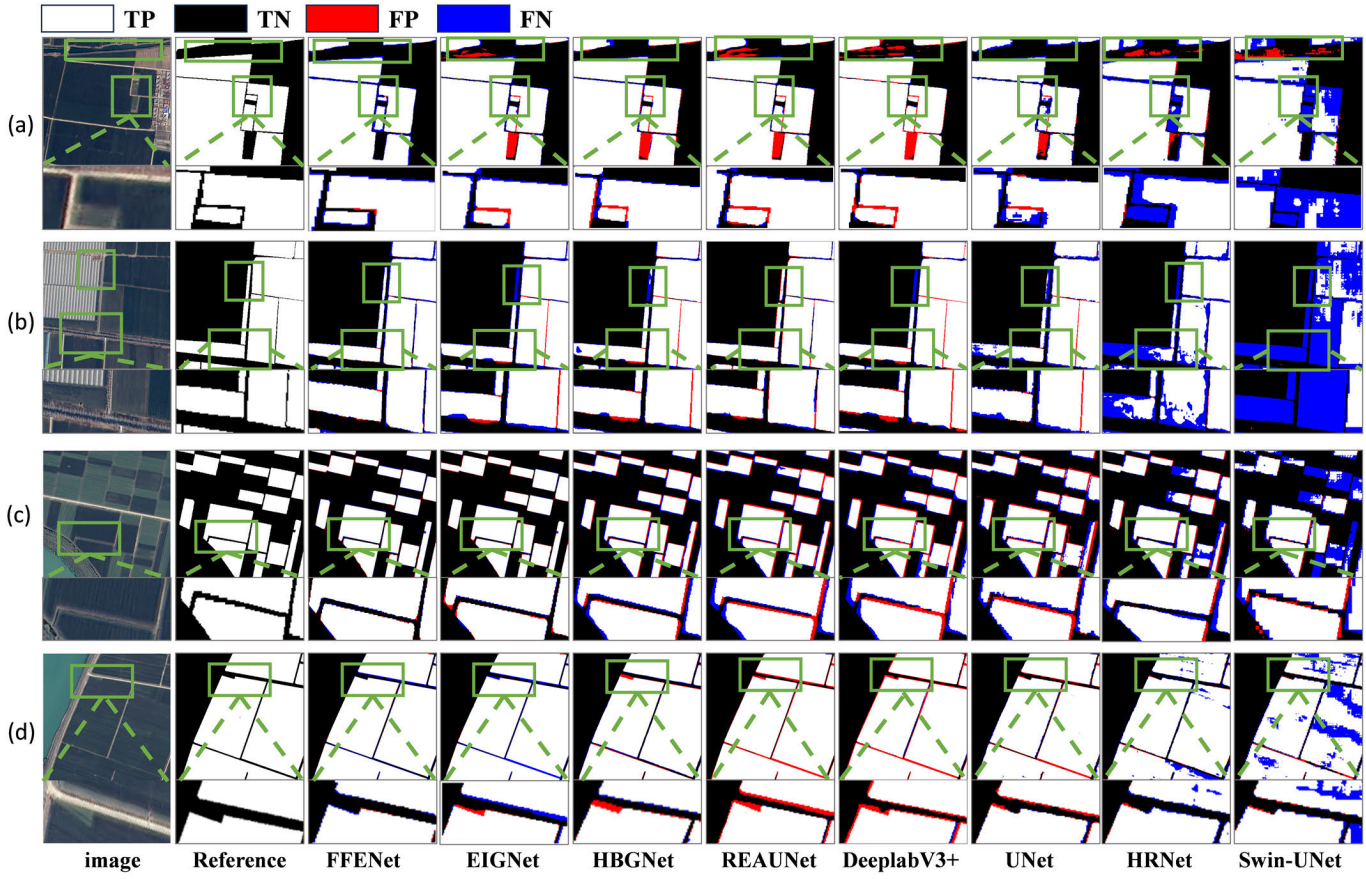


Fig. 8. Farmland plots extracted using different methods. (a)–(d) Comparisons of farmland plots extracted by FFENet and other models. Column 1 presents the remote sensing image patches, column 2 shows the labeled maps, and columns 3–10 display the extraction results of eight models in order: FFENet, EIGNet, HBGNet, REAUNet, DeepLabV3 +, UNet, HRNet, and Swin-UNet.

metrics. These findings demonstrate the clear superiority of FFENet in fine-grained agricultural parcel extraction.

After all the models completed testing, four image patches were randomly selected from the experimental results to visually demonstrate the extraction performance of FFENet and the comparison models. A visual comparison of the field parcel segmentation results in Fig. 8 reveals clear performance differences among the models. HRNet and Swin-UNet exhibited pronounced undersegmentation in field parcel areas, resulting in a high incidence of false negatives. Although UNet and DeepLabV3+ enhanced the completeness of field parcels, their outputs contained notable jagged artifacts along the boundaries. In contrast, EIGNet, HBGNet, and REAUNet not only produced more complete field parcel segmentation but also substantially improved the delineation of fine-scale features, including field ridges and farm roads. Notably, the proposed FFENet achieved the best performance across all the test cases. For large field parcels, it generated smooth boundaries, complete structures, and sharp corners; for small parcels, it delivered more accurate contour extraction while minimizing false negatives. These results validate the effectiveness of the proposed boundary enhancement module in refining field parcel boundaries.

2) Efficiency Analysis: To comprehensively evaluate the practical value of FFENet, we assessed its system efficiency from four aspects: model weight size (disk size), number of parameters, floating-point operations (FLOPs), and prediction time, as shown in Fig. 9. The experimental results showed that compared with the traditional segmentation model UNet, FFENet improved the IoU by 4.7 percentage points while maintaining comparable inference times, achieving a significant accuracy gain. In comparison to the lightweight models Swin-Unet and HRNet (inference times of 0.12 and 0.10 s, parameter sizes of 25.63 and 28.68 MB, and IoUs of 76.50% and 78.03%, respectively), FFENet better met the high-precision requirements for agricultural applications, with an IoU of 89.5%. Relative to edge-optimization-based models, FFENet achieved an IoU 3.72% higher than that of REAUNet while using only 21.9% of its parameters. In contrast, HBGNet attained comparable accuracy to FFENet, but FFENet demonstrated clear advantages in parameter efficiency, FLOPs, and inference speed.

Overall, compared with traditional CNN-based and lightweight models, the proposed approach did not show a significant advantage in terms of parameter count or computational complexity. However, relative to edge-optimization-based models, it achieved reductions in both parameters and computational cost while maintaining higher accuracy, thereby striking a better balance between accuracy and complexity. This combination of strengths enhances its practical utility for real-world agricultural field parcel extraction.

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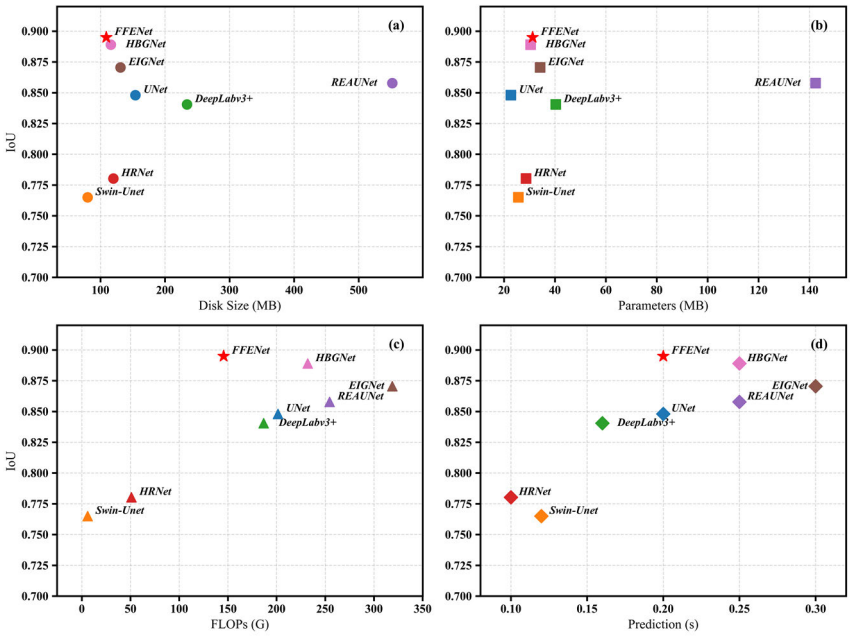


Fig. 9. Model performance in termsof disk size, FLOPs, parameters, and prediction time. (a) Model weight size (disk size) of each model, (b) number of parameters of each model, (c) floating-point operations (FLOPs) of each model, (d) prediction time of each model.

TABLE II
ACCURACY EVALUATION RESULTS OF ABLATION EXPERIMENTS

Baseline	EEM HA-EAM	FFM	EDWC	OA(%)	P (%)	R(%)	F1(%)	IOU(%)
✓				91.53	92.57	91.84	92.20	85.38
✓	✓			92.21	93.30	92.42	92.87	86.45
✓	✓	✓		93.36	94.52	93.63	94.07	88.30
✓	✓	✓	✓	**93.80**	**95.00**	**94.20**	**94.60**	**89.50**

D. Ablation Experiments

To validate the effectiveness and robustness of the HA-EAM, FFM, and EDWC modules for agricultural parcel segmentation in our proposed model, we conducted comprehensive ablation studies on an established farmland parcel dataset. The proposed model was improved based on ResNet18; therefore, in the experiments, it was used as an encoder with layer-by-layer deconvolutional upsampling as a decoder and softmax as a classifier, and a U-Net model was constructed as a baseline.

The experimental results are shown in Fig. 10, indicating that after introducing the EEM and HA-EAM to the baseline, the field integrity showed a significant improvement, and the edge roughness was reduced compared with the baseline network. With the additional incorporation of the FFM module branch, the field segmentation integrity was further enhanced, and the field boundaries appeared more clearly defined. Finally, to validate the impact of the proposed EDWC method on the extraction performance while keeping the feature extractor (i.e., encoder and decoder) unchanged, the softmax classification method was used to process the feature values output from the final layer of the neural network for feature classification. After introducing the EDWC, the field edges and corners demonstrated further refinement, the segmentation completeness reached higher levels, and the resulting field shapes exhibited better consistency with the actual fields.

The comparative results of the performance metrics of the ablation experiments are shown in Table II, to quantitatively analyze the effectiveness of the introduced branches in the model.

As shown in Table II, incorporating EEM and HA-EAM into the baseline model increased the IoU and $F1$-score by 1.07% and 0.67%, respectively. Subsequently, the addition of FFM yielded a marginal improvement in both the metrics. Finally, replacing the softmax classifier with EDWC enabled the model to achieve an OA of 93.8%, precision of 95.0%, recall of 94.20%, $F1$-score of 94.60%, and IoU of 89.5%, thereby further boosting the segmentation accuracy.

E. Model Transferability Experiment

To evaluate the generalization ability of the model in different agricultural scenarios, we selected Linfen City in Shanxi Province and Jingmen City in Hubei Province for comparative validation. Linfen City, a typical representative of the semi-arid Loess Plateau, experiences frequent spring droughts, often accompanied by atmospheric droughts induced by high wind speeds. Its unique environment provides a highly

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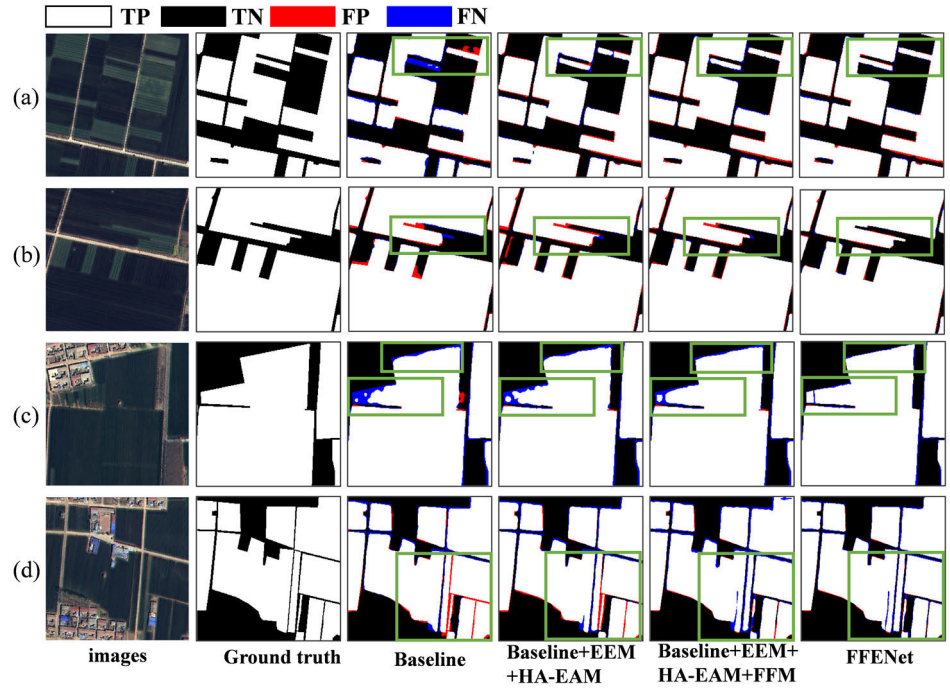


Fig. 10. Results of module ablation experiments for FFENet. (a)–(d) Comparisons of farmland parcel extraction results in ablation experiments.

TABLE III
ACCURACY ASSESSMENT IN THE MODEL TRANSFERABILITY EXPERIMENT

Study area	Method	IOU(%)	F1(%)
LF	ZT	65.32	78.92
	FT	**83.14**	**92.53**
	FLT	81.96	91.43
JM	ZT	59.17	76.54
	FT	**82.46**	**90.53**
	FLT	81.65	89.94

TABLE IV
ACCURACY EVALUATION OF LOSS FUNCTION HYPERPARAMETER ABLATION

(β, γ)	IoU (%)	F1 (%)	Edge Precision (%)
(1.0,0.5)	85.38	92.15	70.23
(1.0,0.8)	88.43	94.03	82.16
(1.0,1.0)	**89.50**	**94.60**	**83.72**
(1.0,1.5)	78.43	88.21	86.87
(1.0,2.0)	60.12	75.18	88.53

differentiated test site to assess model stability under drought conditions. In Jingmen City, winter wheat and winter rapeseed are the main crops, with highly overlapping growth periods of up to seven months, offering an ideal site to verify the model's adaptability in mixed cropping scenarios. Both areas differ significantly from the study area in Taian City in terms of cropping structure and water availability. Together, they constitute a systematic validation framework to comprehensively evaluate the model's generalization performance across different agroecological environments.

We conducted three transfer experiments: zero-shot transfer (ZT), few-sample transfer (FT), and full-sample transfer (FLT). ZT refers to directly testing the model trained in the Taian region on a new region, FT refers to pretraining the model in Taian and then fine-tuning it using a small amount of data from the new region, and FLT refers to training a model using only local data. In FT, 100 samples from the new region were used for 50 training epochs. In FLT, 300 local samples were used for 100 epochs. Table III summarizes the evaluation metrics for the three scenarios.

Fig. 11 presents the extraction visualization results for the two regions. $\text{FFENet}_{\text{FT}}$ achieved the highest accuracy, whereas $\text{FFENet}_{\text{ZT}}$ exhibited the lowest, with a large number of false negatives and false positives. The IoU of $\text{FFENet}_{\text{FT}}$ on the independent test sets of Linfen and Jingmen reached 83.14% and 82.46%, respectively, which were 1.18% and 0.81% higher than those of the directly trained $\text{FFENet}_{\text{FLT}}$, demonstrating the effectiveness of $\text{FFENet}_{\text{FT}}$ for model transfer to new geographic regions.

F. Loss Function Hyperparameter Experimental Analysis

1) Hyperparameter Ablation Study: To ensure optimal model performance and enhance model robustness, a sensitivity analysis of the hyperparameters in the loss function is required. Since β controls the learning of semantic regions within the parcel, testing confirmed that it ensures the effective representation of semantic features. Therefore, this part fixes β and focuses on testing the influence of the edge supervision intensity parameter, γ. Experiments were conducted using γ values of 0.5, 0.8, 1.0, 1.5, and 2.0 on the parcel dataset, and IoU, $F1$, and edge precision values were recorded. The results are shown in Table IV. When $\gamma = 1.0$, the model achieved the

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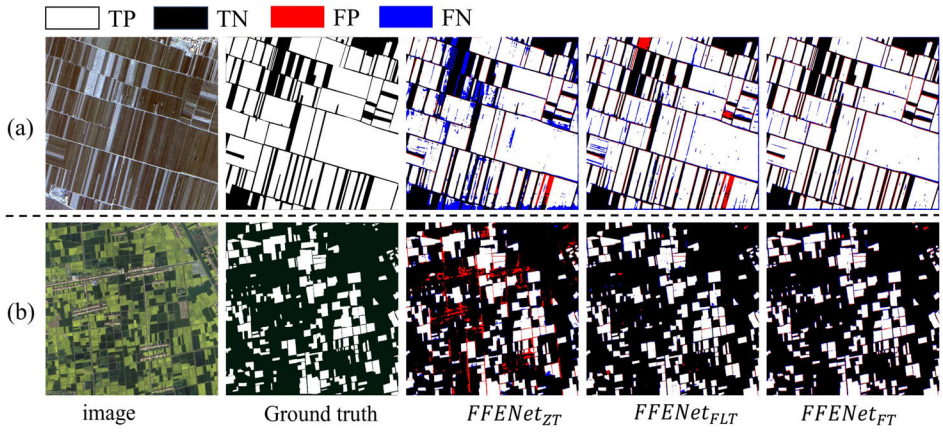


Fig. 11. Model results in the model transferability experiment. (a) Linfen city. (b) Jingmen city.

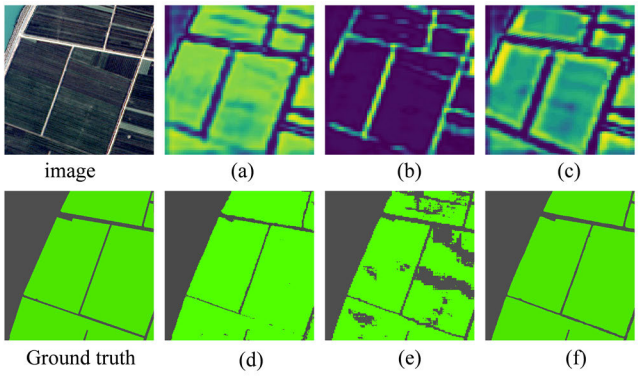


Fig. 12. Feature response maps and segmentation results under different parameters. (a)–(c) Show feature response maps for $\gamma = 0.8$, 1.5 and 1, respectively, (d)–(f) show the corresponding segmentation results.

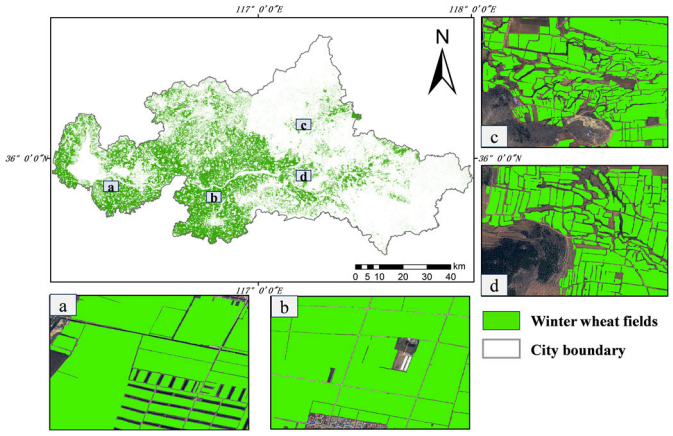


Fig. 13. Farmland plot extraction results in Taian city (exemplified with winter wheat). (a) and (b) Winter wheat parcel extraction results in plain areas, (c) and (d) inter wheat parcel extraction results in hilly areas.

best training results, indicating that the edge weight parameter must balance the ratio between global semantic learning and local boundary optimization to achieve optimal parcel segmentation performance. Therefore, in subsequent experiments, we set the hyperparameters to $\beta = 1.0$ and $\gamma = 1.0$.

2) Analysis of the Action Mechanism of Hyperparameters: To elucidate the action mechanism of hyperparameter γ from the perspective of model learning, we visualized the feature activation maps at the terminal layer of the model decoder and their corresponding segmentation results under different γ values. Overall, the core function of γ lies in regulating the gradient proportion of edge loss and semantic loss during backpropagation, thereby influencing the model's feature learning resource allocation between edge details and regional consistency. Specifically, when the edge supervision weight γ was relatively low (e.g., 0.8), the proportion of edge loss in the total gradient was insufficient, making it difficult for the model to obtain sufficient boundary localization signals during backpropagation and resulting in discrete and discontinuous weak activation of feature responses [Fig. 12(a)] in edge regions. Consequently, the final segmentation results [Fig. 12(d)] exhibited blurred or missing boundaries. Conversely, when γ was excessively high (e.g., 1.5), edge loss dominated the optimization process, resulting in excessive concentration of feature learning resources on boundary extraction at the expense of stable semantic modeling in the internal regions of farmland plots. As shown in Fig. 12(b), feature responses were highly concentrated in the edge zones, while internal activation was significantly weakened, leading to the emergence of holes and impaired connectivity in the segmentation results [Fig. 12(e)]. An optimal balance between edge loss and semantic loss was achieved only when $\gamma = 1.0$. The model was able to receive sufficiently strong gradients for both boundary localization and regional integrity. As shown in Fig. 12(c), continuous and clear response bands were formed in edge regions, while internal regions maintained stable activation. Consequently, the output segmentation results [Fig. 12(f)] exhibited accurate boundaries and intact regions.

G. Farmland Plot Extraction Results

Using the trained FFENet model, this study completed farmland plot mapping for Taian City and Juye County based on GF-2 PMS images via 512×512 pixel block processing. The mapping results are shown in Figs. 13 and 14. Specifically, Fig. 13 presents the farmland plot extraction results of Taian City, where (a) and (b) correspond to the results of plain areas,

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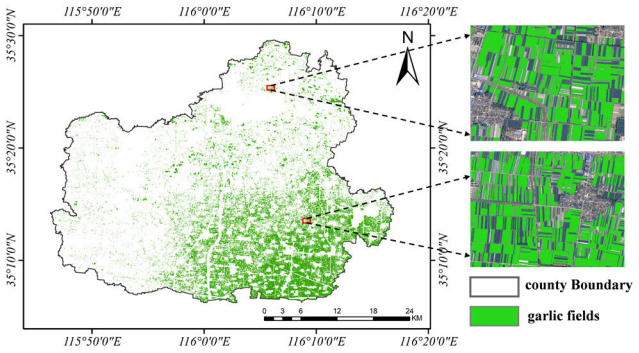


Fig. 14. Farmland plot extraction results in Juye county (exemplified with garlic).

and (c) and (d) represents those of hilly areas; Fig. 14 displays the extraction results of plain areas in Juye County.

From the spatial distribution of the mapping results, the FFENet method effectively distinguished farmland plots from other land-cover types, achieving high-precision foreground–background separation. For both the winter wheat plots in Taian City and the garlic plots in Juye County, the model successfully eliminated interference from field roads, shelterbelts, and other infrastructure. FFENet achieved fine-grained extraction of farmland plots, exhibiting outstanding performance, especially in maintaining plot-shaped integrity and boundary sharpness. These results verified the model's adaptability to the extraction of farmland plots with different crop morphologies.

V. Discussion

A. Comparison to Representative Models

CNNs use local image patches as feature extraction units. However, in farmland plot segmentation, edge and internal pixel differ fundamentally in their feature characteristics. Traditional CNNs often suffer from mis-segmentation when processing farmland plot boundaries, a limitation shared by models such as UNet, Swin-UNet, DeepLabV3+, and HRNet. As shown in Fig. 8, UNet exhibits severe jaggedness in segmentation edges due to its fixed receptive field and simple skip connections. Owing to insufficient global context modeling, Swin-Unet and HRNet often produce incomplete segmentation for large-scale farmland areas. The dilated convolutions in DeepLabV3+ introduce gaps in feature representation, disrupting edge continuity and compromising boundary smoothness.

These edge-enhanced models differ significantly in their core architectural choices, particularly in their edge extraction strategies. EIGNet uses the directional convolution module to extract multilevel spatial details and enhance the perception of complex textures. While HBGNet mines explicit boundary features through Laplacian convolution, it has a strong dependence on large-scale annotated data and is prone to overfitting, leading to performance degradation. In contrast, FFENet adopts an improved Sobel operator for edge extraction, offering lightweight, stable, and highly adaptable performance. The improved Sobel operator mitigates the overfitting risks associated with deep-learning-based edge extraction and efficiently captures reliable edge features from limited samples. While the traditional Sobel operator computes gradients only in the horizontal and vertical directions, making it prone to missing oblique or curved boundaries, FFENet incorporates additional diagonal gradient calculations to better handle complex farmland plot boundaries. FFENet further fuses multidirectional gradient features through convolutional layers and compensates for missing details through residual connections, ensuring robust edge feature representation.

In terms of feature fusion, FFENet introduces an edge weight map and combines it with a channel attention mechanism and residual structure, achieving dynamic weighted fusion of semantic and edge features. In contrast, HBGNet uses the boundary guidance module that uses low-level boundary features to constrain high-level semantic features. While the module can leverage boundary information, it lacks inter-channel correlation modeling, making it prone to the dilution of boundary features by semantic features during the fusion process. Meanwhile, REAUNet relies on the dual attention block and deep supervision to enhance the model's edge perception capability, focusing on weighting in the spatial and channel dimensions of feature maps. In contrast, FFENet achieves a spatially adaptive fusion strategy through the weighted fusion of edge weight maps: it relies more on edge features in boundary regions and semantic features in internal regions. Finally, FFM refines the fusion results through a residual structure integrated with a channel attention mechanism. While the residual path ensures no information loss, the channel attention further emphasizes key channels related to boundaries, textures, and shapes, and suppresses components with heavy noise or irrelevant semantics, achieving dual constraints on the feature space and channel dimension.

B. In-Depth Analysis of the HA-EAM Module

In the process of multiscale edge feature fusion, the semantic differences between different levels of features and the spatial information loss caused by upsampling are key issues affecting boundary quality. To address this, FFENet introduces the HA-EAM module to adaptively enhance edge feature representation via a hybrid attention mechanism. First, this module performs upsampling on the multiscale edge feature maps extracted by EEM, and then executes channel concatenation fusion after standardizing the resolution. To address the noise interference and loss of spatial details caused by direct fusion, HA-EAM introduces a hybrid attention mechanism: the channel attention module dynamically learns channel weights via 1×1 convolutions, adaptively enhancing channels related to boundary gradients and texture variations while suppressing noisy channels that contribute little to boundary discrimination; the spatial attention module constructs a spatial weight map, and by incorporating global contextual relationships, it focuses on enhancing the activation intensity of high-response regions around boundaries. Both the modules aim to improve the feature representation of edge objects. Finally, the feature maps from the two attention branches are concatenated to generate E_p and E_w, Specifically, E_w serves as spatial prior knowledge to further guide the optimization of both the feature fusion and classification modules, thereby enhancing the overall performance of the network.

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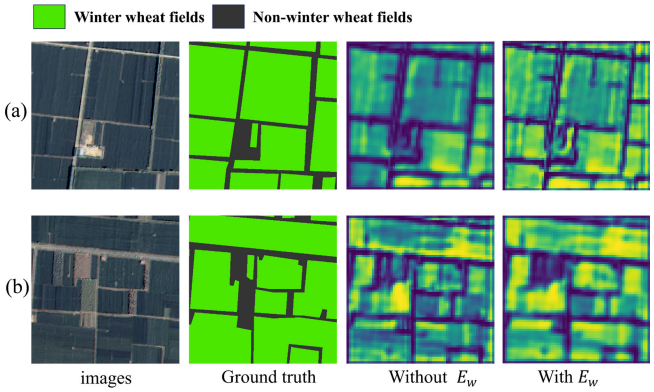


Fig. 15. Visualization example of the output features of the FFM-2 module on Taian dataset images. (a) and (b) Show the model feature response results with and without EW, respectively.

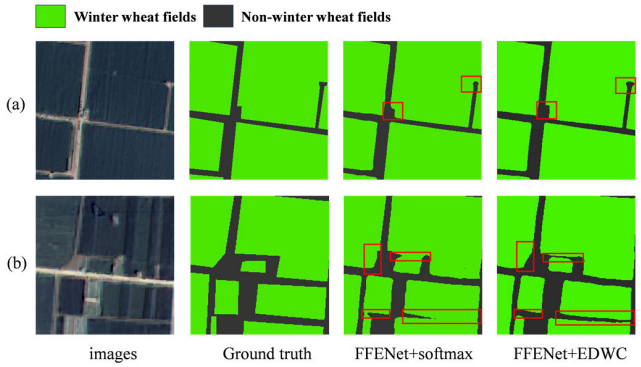


Fig. 16. Ablation experimental results of softmax and EDWC classifiers. (a) and (b) Show the farmland parcel segmentation results of the model with and without the EDWC module, respectively.

TABLE V
ACCURACY RESULTS OF ABLATION STUDY

Method	OA (%)	P (%)	R (%)	F1 (%)	IOU (%)
$\text{FFENet}_{\text{noHA-EAM}}$	92.54	93.21	92.20	93.11	86.10
$\text{FFENet}_{\text{noEDWC}}$	93.36	94.52	93.63	94.07	88.30
FFENet	**93.80**	**95.00**	**94.20**	**94.60**	**89.50**

To verify the effectiveness of HA-EAM in edge information extraction, Fig. 15 presents the visualized feature outputs of the FFM-2 module, where bright regions indicate the areas receiving the model's strongest attention. A comparison between feature fusion results without E_w and those with E_w revealed that when E_w is not introduced, the model can effectively respond to the main areas of farmland but exhibits insufficient attention to edge regions and small-sized fields. After incorporating E_w, the model's responsiveness to edge areas was significantly enhanced, and redundant attention to nonfarmland regions was substantially reduced, allowing edge features in complex backgrounds to be accurately captured. In addition, ablation experiments conducted on the farmland plot dataset (Table V) demonstrated that the introduction of HA-EAM increases the $F1$-score by 1.49% and IoU by 3.4% on the dataset. These results confirm that compared with simply combining high-level and low-level features to enhance edge information, HA-EAM can significantly improve model accuracy and more effectively capture fuzzy boundary features.

C. In-Depth Analysis of the EDWC Module

The EDWC classifier proposed in this study was developed as an improved extension of Softmax, designed to enhance the classification accuracy of edge and corner pixels in farmland plots. Although FFENet improved feature consistency among pixels within the same plot, Softmax computed class probabilities based on all pixel features, causing its predictions to be biased toward the majority of interior pixels. Therefore, misclassification frequently occurred at edges and corners, where feature responses were weak.

To address this limitation, EDWC integrates both semantic and edge feature information from the target pixel and its neighborhood, dynamically adjusting the α value according to the edge density in the edge weight map, dynamic adjustment of the α value: When edge density was high, α value decreased, increasing the influence of edge information; when edge density was low, α value increased, allowing the classifier to rely more on Softmax's inherent semantic discrimination to avoid misclassification. In addition, the EW term introduced an edge-intensity-weighted probability, strengthening the contribution of edge information and producing classification results that more closely matched the actual boundary features.

Fig. 16 shows comparison of the performance of the traditional softmax classifier with that of the proposed EDWC classifier, using the same feature maps. The results obtained with softmax exhibited incomplete edge recognition, misclassification errors, and blurred segmentation, particularly at field corners. In contrast, the EDWC classifier achieved complete and accurate edge detection, producing smoother and clearer boundaries while maintaining sharp corner definitions without misclassification, demonstrating superior alignment with ground truth. These observations confirm that incorporating EDWC improves edge recognition accuracy in the FFENet model, validating the advantages of the proposed classifier design.

D. Limitations

The experimental results demonstrate that the FFENet model can finely extract farmland plot information when processing high-resolution remote sensing images. To comprehensively evaluate the generalization ability of the model on different types of remote sensing datasets, Sentinel-2 images acquired in Taian City during the same period were tested. Sentinel-2 has a spatial resolution of 10 m, which differs significantly from that of GF-2. The results are shown in Fig. 17, where the white regions represent the extracted farmland plot areas and the black regions represent nonfarmland plot areas. Compared with GF-2, winter wheat extraction from Sentinel-2 images was generally less complete, and the edges of the extracted fields were less well-defined.

There are two main reasons for this phenomenon. First, the low resolution of Sentinel-2 images results in insufficient information for critical features, such as the edges and textures of winter wheat plots. This makes it difficult for the model to accurately identify and segment the target areas. Second,

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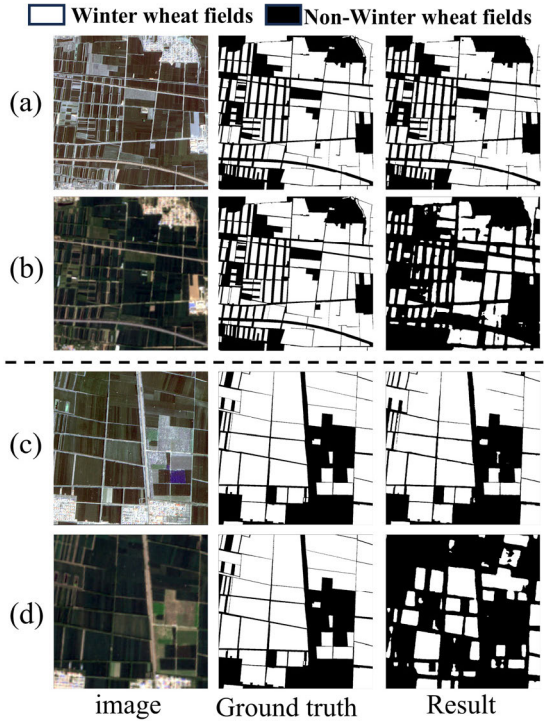


Fig. 17. Extraction results of remote sensing images with different resolutions. (a) and (b) Extraction results of GF-2 and sentinel-2 in the same area and (c) and (d) extraction results of GF-2 and sentinel-2 in the same area.

the model relies heavily on feature representations from high-resolution images, limiting its ability to learn features from medium-resolution remote sensing data. To improve the model's ability to recognize relevant features, future research should focus on enhancing the feature information of medium-resolution images through data augmentation techniques, such as super-resolution reconstruction and image sharpening.

VI. Conclusion

In response to the urgent need for extracting fine-scale farmland plot data from high-resolution remote sensing images, we proposed a refined remote sensing extraction method for farmland plots that integrates semantic and edge features. This method fully incorporates edge features into the entire segmentation process, effectively addressing the problem of frequent errors at farmland plot corners in the extraction results of traditional methods. The experimental results demonstrate that the proposed FFENet model performed excellently in the task of fine-scale farmland plot extraction.

The main contributions of this study are as follows.

1) A pixelwise feature extraction method that integrates edge and semantic features is proposed, considerably enhancing the consistency of features among homogeneous pixels. FFENet effectively captures potential farmland plot boundary information through its edge extraction branch and edge perception module, combined with attention mechanisms and residual structures, while efficiently fusing edge, spatial, and semantic features.
2) A classification method that comprehensively considers both pixel-level features and edge probability is developed, which substantially enhances the classification accuracy for edge and corner pixels, and effectively improves the geometric consistency between the extracted fields and actual field shapes. By improving the traditional softmax classification mechanism through incorporating an edge EW term and dynamically adjusted weighting coefficients (α), FFENet enables the classifier to adaptively enhance edge information influence in boundary regions while maintaining classification balance when edge information is insufficient, thus effectively solving edge misclassification problems without overrelying on edge information.
3) A weighted loss function (Loss) incorporating edge information is proposed, which effectively reduces misjudgment probabilities at the edges by combining binary cross-entropy loss with edge loss, strengthens the constraining role of edge information during training, and significantly improves the farmland plot shape extraction.

Currently, the FFENet model requires meter-level high-resolution remote sensing imagery as the data source to complete the extraction of fine farmland plot data. In our subsequent research, we plan to incorporate remote sensing image super-resolution techniques to meet the demand for extracting fine farmland plots from 10-m level medium-low resolution remote sensing imagery. This enhancement will better enhance the model's applicability to scientific and production practices.

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